Supplemental Table S1. List of differently expressed genes (DEGs) between 'Hongzaosu' and 'Zaosu' pear fruits at 60 DAFB.

Gene ID	Log2 Fold change	Regulated	SwissProt
Pbr015394.1	-1.524	Down	Transcription repressor OFP13
Pbr000537.1	-1.777	Down	Transcription repressor OFP5
Pbr001559.1	2.76	Up	Transcription factor TCP
Pbr013906.1	2.06	Up	Transcription factor TCP19
Pbr031206.1	-4.699	Down	Transcription factor TCP4
Pbr039077.1	2.837	Down	Cyclic dof factor 2 CDF2
Pbr020039.1	-2.764	Down	Dof zinc finger protein DOF58
Pbr029884.1	1.79	Up	Homeobox-leucine zipper protein HAT22
Pbr013533.1	2.44	Up	Homeobox-leucine zipper protein ATHB-7
Pbr024858.1	-2.611	Down	PAP-specific phosphatase HAL2-like
Pbr000136.1	1.231	Up	5'-adenylylsulfate reductase 3, chloroplastic
Pbr031908.1	2.448	Up	5'-adenylylsulfate reductase 3, chloroplastic
Pbr025271.1	2.689	Up	9-cis-epoxycarotenoid dioxygenase NCED1
Pbr039596.1	2.663	Up	9-cis-epoxycarotenoid dioxygenase NCED1
Pbr013248.1	-5.017	Down	Leucoanthocyanidin reductase
Pbr005931.1	-3.446	Down	Bifunctional dihydroflavonol 4-reductase
Pbr020145.1	-4.337	Down	Bifunctional dihydroflavonol 4-reductase
Pbr039986.1	-2.943	Down	Anthocyanidin 3-O-glucosyltransferase 2
Pbr032991.1	-3.8	Down	3-ketoacyl-CoA thiolase 2, peroxisoma
Pbr042806.1	-1.184	Down	3-ketoacyl-CoA thiolase 2, peroxisomal
Pbr008748.1	-1.173	Down	Transcription factor MYB73
Pbr012310.1	-2.825	Down	Transcription factor MYB73-like
Pbr015309.1	-2.481	Down	Transcription factor MYB73-like
Pbr002014.1	-1.723	Down	MYB-related protein 306

Pbr011095.1	-5.049	Down	Transcription factor MYB15
Pbr015230.1	-3.165	Down	Anthocyanin regulatory C1 protein
Pbr018029.1	-4.785	Down	Transcription factor MYB4
Pbr019293.1	-2.177	Down	Transcription factor MYB106
Pbr026080.1	1.57	Up	Transcription factor MYB61
Pbr032528.1	-1.008	Down	MYB-related protein 306
Pbr033541.1	-2.573	Down	MYB-related protein 91
Pbr039674.1	-1.76	Down	Flavonoid 3-O-glucosyltransferase
Pbr019272.1	2.495	Up	Starch synthase 2, chloroplastic/amyloplastic
Pbr001312.1	1.385	Up	Sucrose-phosphatase 1
Pbr005388.1	-2.521	Down	alpha-trehalose-phosphate synthase 7

“Up” means genes up-regulated in the red skinned pear relative to the green skinned pear in different developmental stages, and “Down” means genes down-regulated in the red skinned pear relative to the green skinned pear in different development stages

Supplemental Table S2. RT-qPCR primers used in this study.

Primer name	Primer sequence (5'-3')
FvDFR-F	CACGATTCACGACATTGCGAAATT
FvDFR-R	GAACTCAAACCCCATCTCTTTCAGCTT
FvANS-F	GAAGTGCGTACCCAACTCCATCGT
FvANS-R	ACCTTCTCCTTGTTGACGAGCCC
FvUFGT-F	TTTGGTTCGGTGCTCATA
FvUFGT-R	AGCATCTGTCCCATCTGGT
FvGST-F	CAAGTTCCAGCAATCGAAGA
FvGST-R	TGGGAAGGATCACAAGTTGA
Fv26S -F	AGCCTAACGCAGAGGTTCCAAA
Fv26S -R	GCAGCCCACATTGAAGGGTCTATAGT
FvActin-F	GCCAACCGTGAGAAGATG
FvActin-R	TCCAGAGTCAAGAACAATACCAG
PyMYB114-F	GCCACATCCGTCATAAGACCTC
PyMYB114-R	GCCACTCATGTGTAACCCTTC
PyMYB10-F	GACCAATGTGATAAGACCTCAGCC
PyMYB10-R	CCGTTCTTTGTTGACGACGAC
PybHLH3-F	TTGTGGAGGGAAGTGGCGGT
PybHLH3-R	AGCTCCCTAAGTGTTTGCATCAC
PyDFR-F	GGTTTCATCGGCTCTTGGC
PyDFR-R	CCTTCTTCTGATTCGTGGGGT
PyUFGT-F	CCTTCCCTTTTGCCACTCA
PyUFGT-R	CAGCCACATCGTACACCCTTA
PyANS-F	GAGCAGAAGGAGAAGTAT
PyANS-R	ACAGTGGAAGAAGTAGTC
PyWRKY26-F	GCTTCTTGACTCTCCTGTTCTC
PyWRKY26-R	GTGGTTCTTGCTCTCCTGTT
PyWRKY31-F	GACCTCACCACATCTCCATTAC
PyWRKY31-R	GGGTTGCCCTGGGAAATTA
PyGST-F	TGGGCAAGTTCCAGTTGTAG
PyGST-R	CCTCCAGAGTTGTTCCCAATAG
PyABC-F	CCTTCGAGGCCAGATGTTATT
PyABC-R	AACAGTGCTCTTCCCTGAAC
PyAVP1-F	CTCTTGTGTCCCTTGCTCTATT
PyAVP1-R	CGCCCACTAACAAACCAATAATC
PyAVP2-F	CTCGTTCCTCTTCACGGAATAC
PyAVP2-R	ATGCACGGCTACTTGTACTAAA
PyMYB73-F	GATGGAGTTTGACCCGTTGA
PyMYB73-R	TGGCAATCACATCCCTCATC

Primer name	**Primer sequence (5'-3')**
PyMYB6-F	CTTCACTGTCTCTGTCACTTCC
PyMYB6-R	GCAACTCCGAGGGCATATT
Pbr012310.1-F	GCTCGTCATCTACTTCGTCTTC
Pbr012310.1-R	TGATTCGTTCGGGCTTGTT
Pbr015309.1-F	GACCGAAACGCTATCAAGAA
Pbr015309.1-R	CATGGAGAACATGACCATCGTA
PyERF3-F	GAAGAAACTGGTGAGAGGAAGAG
PyERF3-R	CTTTGTGTGGATCGCGTATTTC
PyERF4.1-F	AAGGCCAAGACCAACTTCC
PyERF4.1-R	CTGATTGTTGCTTCCGCTACTA
PyERF4.2-F	CATCCAACGAGACCCGATTC
PyERF4.2-R	TCTTCTTCCCGGGATCTCTAAT
PyTubulin-F	GGGCTTTGCTCCTCTTACTT
PyTubulin-R	GATCAGCAGCACACATCATATTC
PyActin-F	GACCACCACTGGTCATCTTATC
PyActin-R	CATCTCAGCAGCTTCCTTCTC

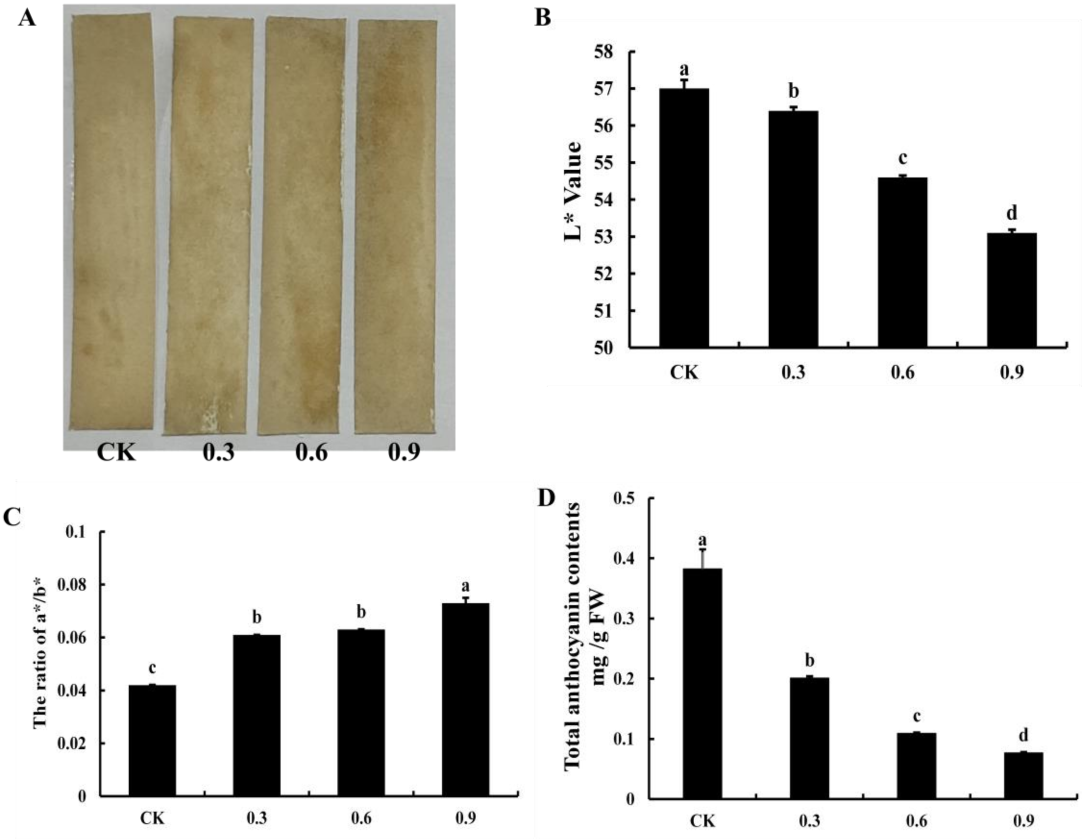
Supplemental Table S3. The list of primers used in this study for transient expression and prokaryotic expression.

Primer name	**Primer sequence (5'-3')**
PyMYB73-277-*EcoR* I-F	ACTAGTGGATCCAAAGAATTCATGGAAGCATTGAATATTTGCTCG
PyMYB73-277-*Xba* I-R	TCATTAAAGCAGGACTCTAGATCAAAGAAGCCCCGAAGGC
PyMYB6-277-*EcoR* I-F	ACTAGTGGATCCAAAGAATTCATGGCGGTGATCAGGAAAGA
PyMYB6-277-*Xba* I-R	TCATTAAAGCAGGACTCTAGATCATCGTAATTGCGAAAATGTCC
PyMYB10C194A-277-*EcoR* I-F	ATACTTTTGAAAGGGCTGCAGCGCCCAGCATTGAGTTAGAGGA
PyMYB10C194A-277-*Xba* I-R	TCCTCTAACTCAATGCTGGGCGCTGCAGCCCTTTCAAAAGTAT
PyMYB10C218A-277-*EcoR* I-F	ATCGACTGTCGGCAAGATCAGCGGCCAATTTTCCTGAAGAAGG
PyMYB10C218A-277-*Xba* I-R	CCTTCTTCAGGAAAATTGGCCGCTGATCTTGCCGACAGTCGAT
PyANS-0800-*Hind* III-F	GTCGACGGTATCGATAAGCTTACAAATGCAAATTCACACACAAAA
PyANS-0800-*Bam*H I-R	CGCTCTAGAACTAGTGGATCCTTTTGGAGCTGGCTTTCGAC
PyDFR-0800-*Hind* III-F	GTCGACGGTATCGATAAGCTTAAATTTCCTAAAAAATATACTCGTGTTGA
PyDFR-0800-*Bam*H I-R	CGCTCTAGAACTAGTGGATCCCTTATAAAATATTTATAGTGGGGGGCT
PyUFGT-0800-*Hind* III-F	GTCGACGGTATCGATAAGCTTTACAAATATTCAGAGCATATATATCAATAGTAGTG
PyUFGT-0800-*Bam*H I-R	CGCTCTAGAACTAGTGGATCCTGACTGACGGTTGCTATAGGTATTAA
Nluc-PyMYB73-*Bam*H I-F	CGAGCTCGGTACCCGGGATCCATGGAAGCATTGAATA
Nluc-PyMYB73-R	CGCGTACGAGATCTGGTCGACAAGAAGCCCCGAA
Nluc-PyMYB6-*Bam*H I-F	CGAGCTCGGTACCCGGGATCCATGGCGGTGATCAGGA
Nluc-PyMYB6-*Sal* I-R	CGCGTACGAGATCTGGTCGACTCATCGTAATTGCG
Cluc-PyMYB10-*Bam*H I-F	CCGGGGCGGTACCCGGGATCCATGGAGGGATAT
Cluc-PyMYB10-*Sal* I-R	ACGAAAGCTCTGCAGGTCGACCTATTCTTCTTTTGA
Cluc-PyMYB73-*Bam*H I-F	CCGGGGCGGTACCCGGGATCCATGGAAGCATTGAAT
Cluc-PyMYB73-*Sal* I-R	ACGAAAGCTCTGCAGGTCGACAAGAAGCCCCGAAGG
PyMYB10-MBP-*Bam*H I-F	AGGATTTCAGAATTCGGATCCATGGAGGGATATAACGTTAACTTGAG
PyMYB10-MBP-*Hind* III-R	ACGACGGCCAGTGCCAAGCTTCTATTCTTCTTTTGAATGATTCCAAAG
PyMYB73-HIS-*Bam*H I-F	CTCGGTACCCTCGAGGGATCCATGGAAGCATTGAATATTTGCTCG
PyMYB73-HIS-*Xba* I-R	AGCAGAGATTACCTATCTAGATCAAAGAAGCCCCGAAGGC
PyMYB6-GST-*EcoR* I-F	CCGCGTGGATCCCCGGAATTCATGGCGGTGATCAGGAAAGA
PyMYB6-GST-*Xho* I-R	GTCACGATGCGGCCGCTCGAGTCATCGTAATTGCGAAAATGTCC

Supplemental Table S4. The list of primers used in this study for Y2H.

Name	Primer Sequences (5'-3')
BK-PyMYB73-*Nde* I-F1	TCAGAGGAGGACCTGCATATGATGGAAGCATTGAATATTTGCTCG
BK-PyMYB73-*Pst* I-R1	CTAGTTATGCGGCCGCTGCAGTGCGGCCAAGATGGTCTCG
BK-PyMYB73-*Nde* I-F1	TCAGAGGAGGACCTGCATATGATGGAAGCATTGAATATTTGCTCG
BK-PyMYB73-*Pst* I-R2	CTAGTTATGCGGCCGCTGCAGATTAACCAAAGATCCGGAAACCG
BK-PyMYB73-*Nde* I-F1	TCAGAGGAGGACCTGCATATGATGGAAGCATTGAATATTTGCTCG
BK-PyMYB73-*Pst* I-R3	CTAGTTATGCGGCCGCTGCAGTCAAAGAAGCCCCGAAGGC
BK-PyMYB73-*Nde* I-F2	TCAGAGGAGGACCTGCATATGGGGTCGATGGAGTTTGACCC
BK-PyMYB73-*Pst* I-R3	CTAGTTATGCGGCCGCTGCAGTCAAAGAAGCCCCGAAGGC
BK-PyMYB73-*Nde* I-F3	TCAGAGGAGGACCTGCATATGCAGGCCCGGTTCGGAAAC
BK-PyMYB73-*Pst* I-R3	CTAGTTATGCGGCCGCTGCAGTCAAAGAAGCCCCGAAGGC
BK-PyMYB6-*Nde* I-F1	TCAGAGGAGGACCTGCATATGATGGCGGTGATCAGGAAAGA
BK-PyMYB6-*Pst* I-R1	CTAGTTATGCGGCCGCTGCAGCTGCGGCGACAGCTGGTT
BK-PyMYB6-*Nde* I-F1	TCAGAGGAGGACCTGCATATGATGGCGGTGATCAGGAAAGA
BK-PyMYB6-*Pst* I-R2	CTAGTTATGCGGCCGCTGCAGACCGCCGTCAGAGCATTTT
BK-PyMYB6-*Nde* I-F1	TCAGAGGAGGACCTGCATATGATGGCGGTGATCAGGAAAGA
BK-PyMYB6-*Pst* I-R3	CTAGTTATGCGGCCGCTGCAGGTCGACCCCGGGAAGTGA
BK-PyMYB6-*Nde* I-F1	TCAGAGGAGGACCTGCATATGATGGCGGTGATCAGGAAAGA
BK-PyMYB6-*Pst* I-R4	CTAGTTATGCGGCCGCTGCAGTCATCGTAATTGCGAAAATGTCC
BK-PyMYB6-*Nde* I-F2	TCAGAGGAGGACCTGCATATGGCCGGGGAGGTGTCGAAC
BK-MYB6-*Pst* -R4	CTAGTTATGCGGCCGCTGCAGTCATCGTAATTGCGAAAATGTCC
BK-PyMYB6-*Nde* I-F3	TCAGAGGAGGACCTGCATATGGGCGTCGTTTTGAATGGAGG
BK-PyMYB6-*Pst* I-R4	CTAGTTATGCGGCCGCTGCAGTCATCGTAATTGCGAAAATGTCC
BK-PyMYB6-*Nde* I-F4	TCAGAGGAGGACCTGCATATGGTGGAGCACCGCGCCTTC
BK-PyMYB6-*Pst* I-R4	CTAGTTATGCGGCCGCTGCAGTCATCGTAATTGCGAAAATGTCC
AD-PyMYB73-*EcoR* I-F	GCCATGGAGGCCAGTGAATTCATGGAAGCATTGAATATTTGCTCG
AD-PyMYB73-*Xho* I-R	ACGATTCATCTGCAGCTCGAGTCAAAGAAGCCCCGAAGGC
AD-PyMYB10-*EcoR* I-F	GCCATGGAGGCCAGTGAATTCATGGAGGGATATAACGTTAACTTGAG
AD-PyMYB10-*Xho* I-R	ACGATTCATCTGCAGCTCGAGCTATTCTTCTTTTGAATGATTCCAAAG
AD-PyMYB114-*EcoR* I-F	GCCATGGAGGCCAGTGAATTCATGAGGAAGGGTGCCTGGAC
AD-PyMYB114-*Xho* I-R	ACGATTCATCTGCAGCTCGAGCTAAATCTTAGTTATCTCTTCTTCTAGATTCCA
AD-PybHLH3-*EcoR* I-F	GCCATGGAGGCCAGTGAATTCATGGCTGCACCGCCGCCA
AD-PybHLH3-*Xho* I-R	ACGATTCATCTGCAGCTCGAGTTAAGAGTCAGATTGGGGTATAATTTG

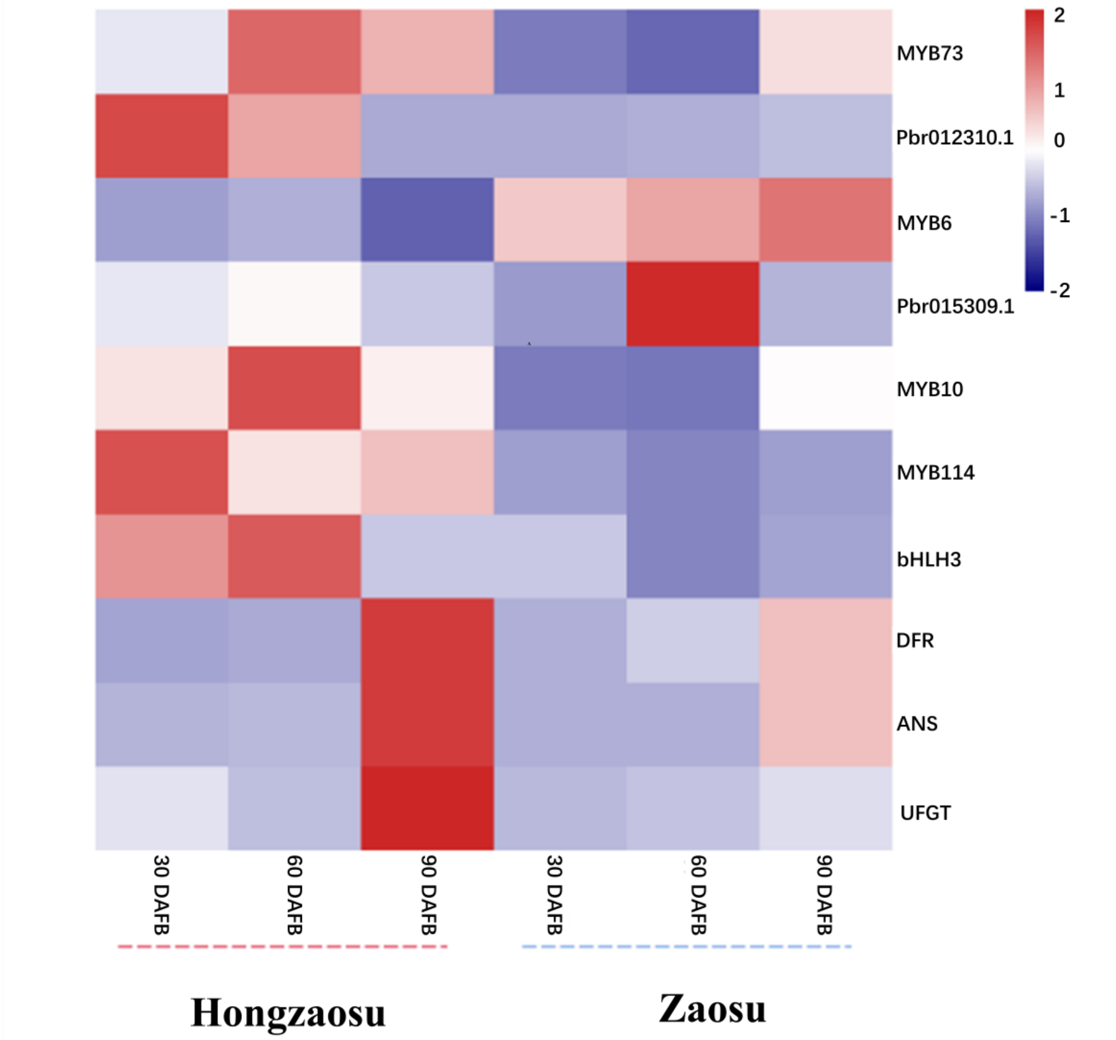
Supplemental Figure 1



Supplemental Figure 1. Production of endogenous H_2S during the pear fruit development.

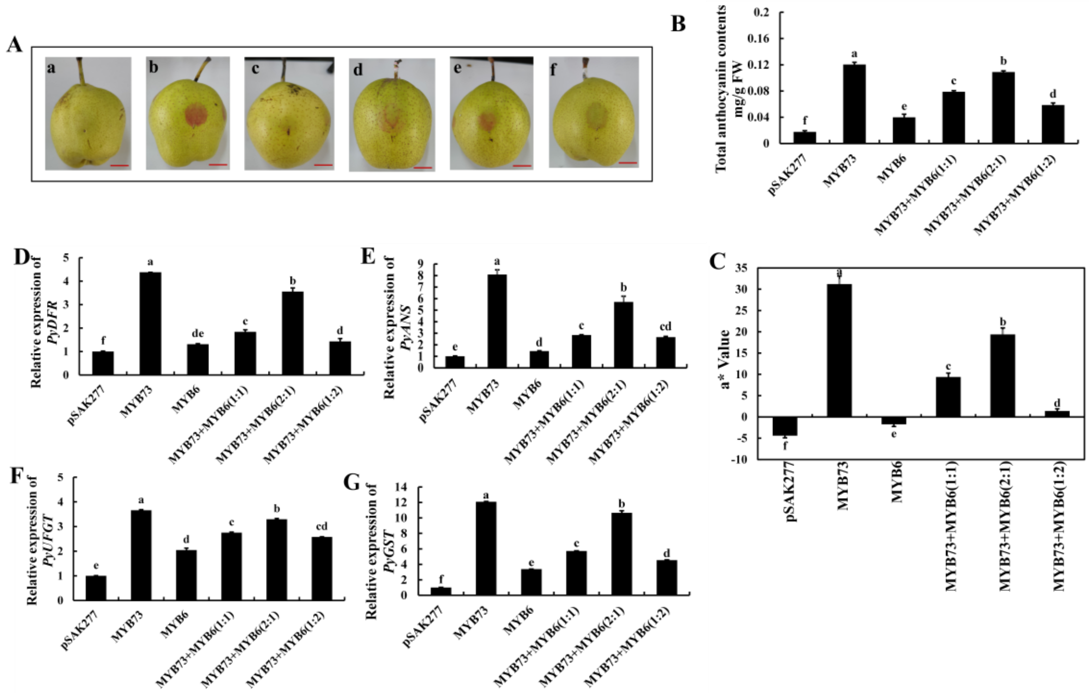
(A) The endogenous H_2S production of the nearly mature bagged 'Aohong No.1' peels which treated with different H_2S concentrations by lead sulfide method. **(B)-(C)** Representing the change of L* value and the ratio of a*/b* corresponding to **(A)**. "L*" represents the brightness : 0~100 means from black to white；"a*" represents the range from red to green；"b*" represents the range from yellow to blue. And lower the L* value and higher ratio of a */ b* represented that the more H_2S was absorbed on the lead acetate test filter paper. Bars indicate mean values ± SD from six biological replicates. **(D)** Total anthocyanin contents were measured the nearly mature bagged 'Aohong No.1' peels which was treated with different H_2S concentrations. Bars indicate mean values ± SD from six biological replicates. Statistical analysis was carried out with Student's *t*-test, and significance was marked with different letters ($P < 0.05$).

Supplemental Figure 2



Supplemental Figure 2. Heatmap was used to analysis the expression levels of genes related to anthocyanin metabolism and regulation in 'Hongzaosu' and 'Zaosu' pears at 30, 60, and 90 DAFB, respectively. DAFB, days after full bloom.

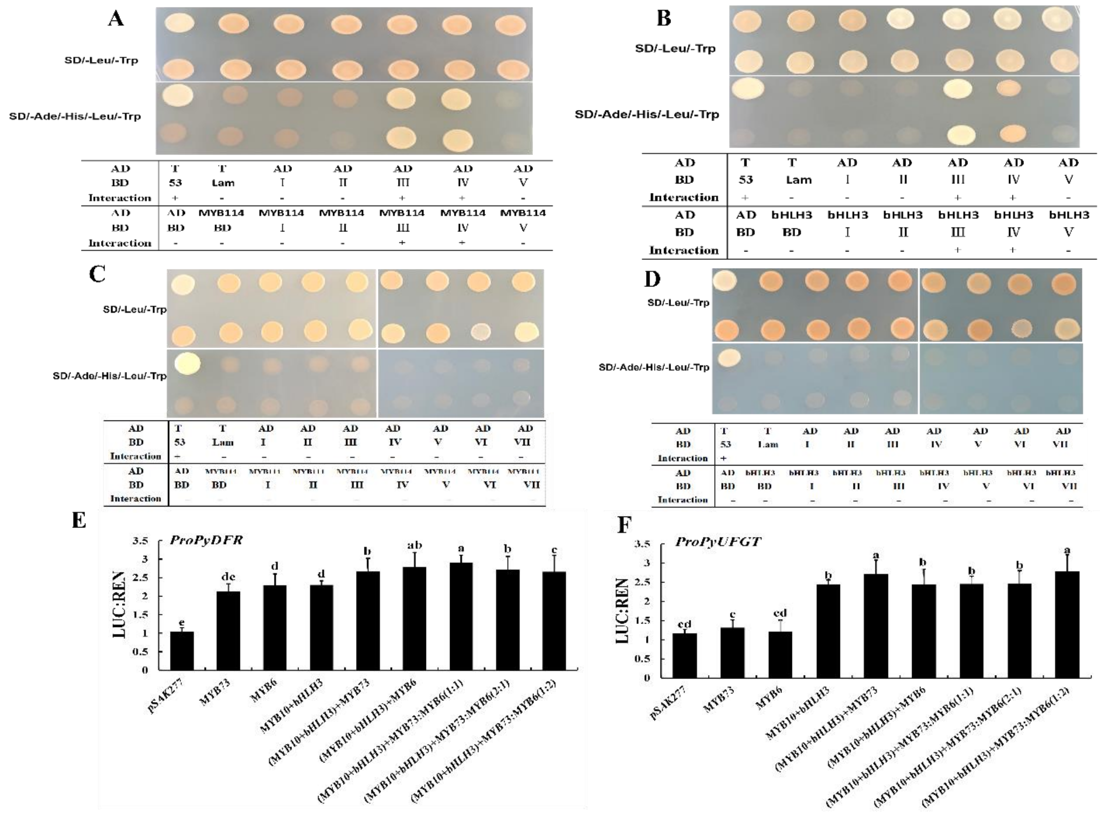
Supplemental Figure 3



Supplemental Figure 3. Co-injection of *PyMYB73/PyMYB6* by transient expression assay in pear skins.

(A) The phenotype changes of 'Zaosu' pears at 7 days after infiltration; a, pSAK277; b, *PyMYB73*; c, *PyMYB6*; d, *PyMYB73+PyMYB6(1:1)*; e, *PyMYB73+PyMYB6(2:1)*; f, *PyMYB73+PyMYB6*(1:2). The scale bars in the figure represents 2 cm. **(B)** The total anthocyanin contents were determined in pear peels. Error bars represent mean values ± SD from five biological repeats. **(C)** Determination of color changed in pear peels by chromatism analysis. "a" represents the range from red to green; Bars represent mean values ± SD from six biological replicates. **(D)-(G)** The relative expression levels of *PyDFR*, *PyANS*, *PyUFGT*, *PyGST* were analyzed by RT-qPCR. Data are presented as means ± SD (n = 3). Statistical analysis was carried out with Student's *t*-test, and significance was marked with different letters ($P < 0.05$).

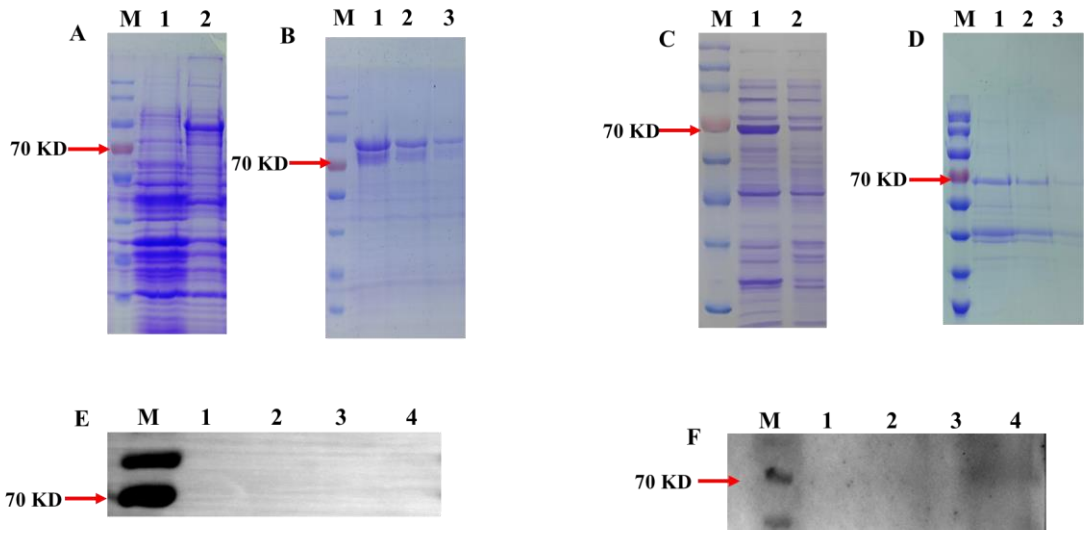
Supplemental Figure 4



Supplemental Figure 4. **Verifying the regulatory pattern of PyMYB73/PyMYB6, PyMYB10 in vivo. (A)** Validation of the interaction of PyMYB73 and PyMYB114 in vivo. **(B)** Validation of the interaction of PyMYB73 and PybHLH3 in vivo. **(C)** Validation of the interaction of PyMYB6 and PyMYB114 in vivo. **(D)** Validation of the interaction of PyMYB6 and PybHLH3 in vivo. **(E)-(F)** The dual luciferase reporter (DLR) assays to verify PyMYB10, PybHLH3 and PyMYB73/PyMYB6 co-transformation activates *PyDFR*, *PyUFGT* promoters, the LUC/REN value was used to express promoter activity. Data are presented as means ± SD (n = 3). Statistical analysis was carried out with Student's *t*-test, and significance was marked with different letters

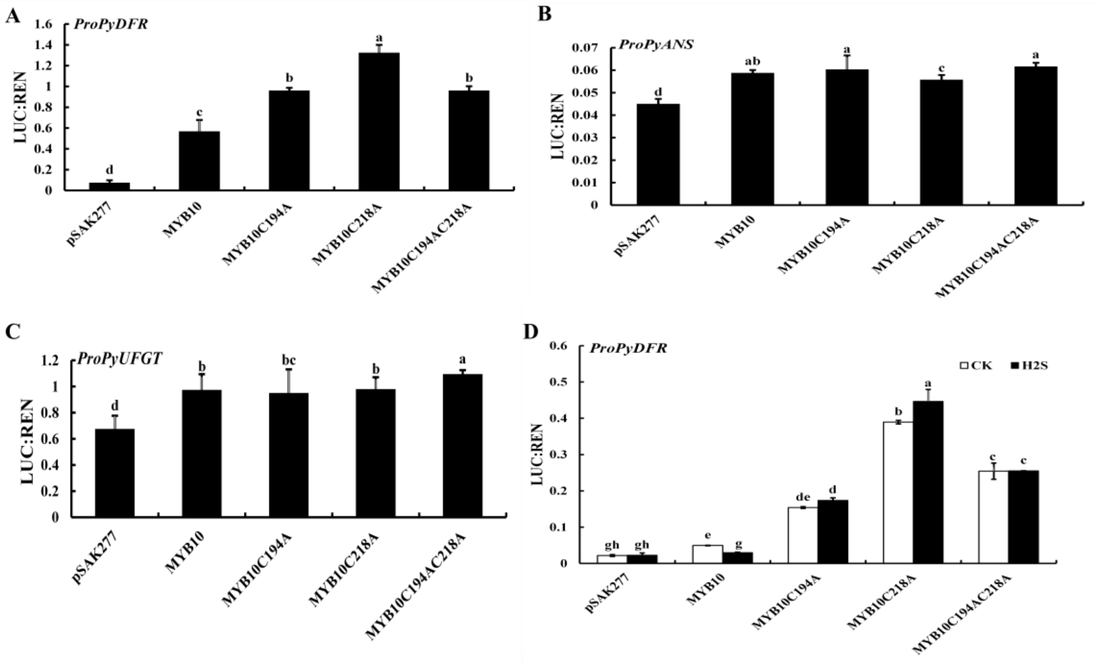
($P < 0.05$).

Supplemental Figure 5



Supplemental Figure 5. PyMYB73/PyMYB6 persulfidation-site and persulfidated protein assays. **(A)** The expression of MYB73. M, maker of protein molecular weight ;1, before induction ;2, after induction. **(B)** Purification of MYB73.M, maker of protein molecular weight;1,2,3, indicates the number of the protein elutions. **(C) -(D)** The expression and purification of PyMYB6. **(E)-(F)** In vitro NaHS-induced persulfidation of PyMYB73**(E)** /PyMYB6**(F)** was detected by using the biotin-switch assay. 1,2,3 represent with 0,400,1000μM NaHS,4 represent 400μM+1mM DDT treatment. Protein was incubated with the indicated concentration of NaHS or NaHS plus DTT (1 mM) for 30 min.

Supplemental Figure 6



Supplemental Figure 6. Dual luciferase reporter (DLR) assays verify that *PyMYB10*, with the mutated thiolated modification site, activates the *PyANS*, *PyDFR*, and *PyUFGT* promoters.

The dual luciferase reporter (DLR) assays verified that after the mutation of *PyMYB10* two cysteine induces the activity of the *PyDFR* **(A),** *PyANS* **(B)**, *PyUFGT***(C)** promoters. Error bars represent the mean values ± SD from three replicate reactions. **(D)** Determination of activation activity to the *PyDFR* promoter with 0.5 mM NaHS treatment for 1 h. The LUC/REN value was used to express promoter activity. Error bars represent the mean values ± SD from three replicate reactions. Statistical analysis was carried out with Student's *t*-test or Duncan's multiple range tests, bars with different letters are significantly different at the $P < 0.05$ level.